VENDORGATE

Vendor Onboarding and Purchase Order Portal

Business Requirements Document (BRD)

Case Study No. 88

Document Attribute	Details
Document Type	Business Requirements Document (BRD)
Project	VendorGate – Vendor Onboarding and Purchase Order Portal
Case Study	Case Study No. 88
Organization Context	Mid-size pharmaceutical distributor
Vendor Base	850 vendors
Vendor Registration	11 documents sent by email
Current PO Process	Purchase orders raised in Excel and approved on paper
Audit Finding	34 purchase orders approved above the approver's limit last year
Version	1.0
Status	Draft / Academic Project Submission

1. Executive Summary

VendorGate is a proposed Vendor Onboarding and Purchase Order Portal for a mid-size pharmaceutical distributor that works with 850 vendors. The current process relies on email for vendor registration documents and on Excel and paper for purchase-order creation and approval.

The audit team found 34 purchase orders approved above the approver's limit last year. The case therefore requires a centralized portal that supports vendor registration, document verification, purchase-order creation with approval limits, and vendor acknowledgement.

The project also addresses governance because Procurement, Finance and Audit each want to own the system. Clear Sponsor, System Owner and Approver responsibilities are therefore required.

2. Business Background

The organization is a mid-size pharmaceutical distributor working with 850 vendors. Vendor registration currently involves 11 documents sent by email. Purchase orders are raised in Excel and approved on paper.

These fragmented processes create a need for a centralized portal where vendor information, documents, purchase orders, approval decisions and acknowledgement activities can be managed and traced.

3. Problem Statement

- Vendor registration depends on 11 documents being exchanged through email.
- Purchase orders are raised in Excel instead of a centralized purchase-order workflow.
- Purchase orders are approved on paper, making approval control and tracking difficult.
- The audit team found 34 purchase orders approved above the approver's limit last year.
- Vendor, document and purchase-order information is not managed through one centralized portal.
- Procurement, Finance and Audit have an ownership conflict regarding the proposed system.

4. Business Need

The organization needs a centralized VendorGate portal that digitizes the vendor onboarding and purchase-order workflow and introduces controlled approval routing.

The intended high-level workflow is:

Vendor Registration → Document Submission → Document Verification → Vendor Approval → Purchase Order Creation → Approval-Limit Routing → PO Approval → Vendor Acknowledgement → Audit Record

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5. Project Objectives

Requirements Engineering: Elicit requirements from vendors, Procurement, Finance and Audit through document analysis and a stakeholder workshop; define functional and non-functional requirements and prioritize them using MoSCoW.

Software Design & UML: Produce consistent use case, class, sequence, activity and purchase-order state diagrams, with a written cohesion and coupling justification.

Testing & Quality Assurance: Design Boundary Value Analysis tests, equivalence classes and a decision table for approval-limit routing, and calculate DRE and defect density from the test log.

Stakeholders & Communication: Resolve the ownership conflict by defining Sponsor, System Owner and Approver roles, classify stakeholders by interest and influence, and define communication needs.

Scheduling & Milestones: Complete dependency analysis, demonstrate the effect of policy-sign-off delays, identify the critical path and float, and produce a Gantt chart with milestones.

Risk Management: Build a risk register using probability × impact exposure, rank risks, define responses and owners, and keep occurred problems in a separate issue log.

6. Stakeholders

Stakeholder	Role / Interest	Primary Concern
Vendors	External users of VendorGate	Registration, documents and PO acknowledgement
Procurement	Operational business stakeholder	Vendor onboarding and PO creation
Finance	Financial control stakeholder	Approval limits and financial authorization
Audit	Independent control stakeholder	Auditability, traceability and approval compliance

The case study explicitly identifies an ownership conflict because Procurement, Finance and Audit each want to be the owner of the system.

7. Proposed Governance Structure

To resolve the ownership conflict, VendorGate should distinguish between sponsorship, system ownership and approval authority. The following is a proposed governance arrangement for the academic project and should be confirmed through stakeholder discussion.

Role	Proposed Responsibility	Justification
Sponsor	Provides overall business direction and resolves cross-department conflicts.	Provides executive-level direction without taking over day-to-day system ownership.
System Owner	Procurement.	The core operational scope is vendor onboarding and purchase-order processing.
Approver / Approval Authority	Finance and designated approval-level users.	Approval limits are financial authorization controls.
Audit / Control Function	Audit.	Provides independent oversight of transactions and audit records.

Vendor	External system user.	Provides registration information, documents and PO acknowledgement.

8. Current / As-Is Process

8.1 Vendor Registration

Vendor → Prepare 11 documents → Send documents by email → Manual handling/verification → Registration

8.2 Purchase Order Process

Purchase Order raised → Excel → Paper approval → Vendor

The case study's audit finding of 34 above-limit approvals demonstrates the control weakness that the new system must address.

9. Proposed / To-Be Process

Vendor → Registration → Document Submission → Verification → Vendor Approval → PO Creation → Approval-Limit Check → Correct Approval Level → PO Approval → Vendor Acknowledgement → Audit Record

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10. High-Level Business Requirements

ID	Business Requirement
BR-01	The system shall provide a centralized portal for vendor registration.
BR-02	The system shall support submission of the required vendor registration documentation.
BR-03	The system shall support vendor document verification.
BR-04	The system shall maintain vendor registration information digitally.
BR-05	The system shall provide a centralized mechanism for purchase-order creation.
BR-06	The system shall apply defined approval limits to purchase orders.
BR-07	The system shall route purchase orders according to the applicable approval level.
BR-08	The system shall support authorized purchase-order approval.
BR-09	The system shall allow vendors to acknowledge purchase orders.

BR-10	The system shall maintain records required for auditability and traceability.
BR-11	The system shall provide appropriate controls for Procurement, Finance, Audit and Vendor activities.
BR-12	The system shall support measurable requirements for security, availability and auditability.

11. Approval-Limit Business Rules

Approval Level	Purchase Order Amount
Level 1	Up to ₹50,000
Level 2	Above ₹50,000 up to ₹2,00,000
Level 3	Above ₹2,00,000 up to ₹10,00,000
Level 4	Above ₹10,00,000

A purchase order must be routed to an authorized approver according to its monetary value and the applicable approval limit. The detailed boundary values and decision table will be documented in the Test Plan.

12. Business Rules

- Vendor registration shall be performed through the VendorGate portal.
- Vendor registration is based on the existing requirement of 11 documents.
- Submitted vendor documentation shall be subject to verification.
- Purchase orders shall be created through VendorGate instead of relying on Excel as the primary process.
- Purchase orders shall be routed according to the four approval levels defined in the case study.
- Only an appropriately authorized approver should approve a purchase order.
- Vendors shall be able to acknowledge purchase orders through the portal.
- Relevant vendor and purchase-order activities shall be traceable for audit purposes.

13. Scope

13.1 In Scope

- Vendor registration
- Vendor information capture
- Vendor document submission
- Document verification
- Vendor onboarding
- Purchase-order creation
- Approval-limit routing
- Purchase-order approval

- Vendor acknowledgement
- Auditability and traceability
- Security and availability requirements
- Stakeholder governance and communication
- Testing of approval-limit routing

13.2 Out of Scope / Not Specified by the Case

- Payment processing
- Inventory management
- Invoice management
- Shipment tracking
- Accounting/ERP integration
- Mobile application
- AI-based vendor assessment
- Other features not specified by the case study unless later approved through change control

14. Business Objectives to Requirement Mapping

Objective	Related Requirement(s)
Centralize vendor registration	BR-01
Digitize document submission	BR-02
Improve document control	BR-03
Centralize vendor records	BR-04
Centralize PO creation	BR-05
Prevent approval-limit violations	BR-06, BR-07
Control PO approvals	BR-08
Enable vendor acknowledgement	BR-09
Improve auditability	BR-10
Establish departmental controls	BR-11
Address measurable security, availability and auditability needs	BR-12

15. Non-Functional Business Expectations

The case study requires the final SRS to contain eight measurable non-functional requirements, covering auditability, security and availability. At BRD level these are business expectations; the exact measurable targets will be defined in the SRS.

- Auditability: relevant vendor and purchase-order activities should be traceable.
- Security: users should only perform activities appropriate to their role.
- Availability: the portal should be available for its intended business users.
- Measurability: the final SRS shall convert these expectations into measurable targets.

16. Assumptions

ID	Assumption
A-01	VendorGate will be used by the organization's vendors and internal departments.
A-02	Vendors will interact with the system for registration and PO acknowledgement.
A-03	Procurement, Finance and Audit will participate in requirements and governance activities.
A-04	The 11 existing vendor-registration documents form the basis of the registration requirement.
A-05	The four approval levels provided in the case study form the basis for PO approval routing.
A-06	Stakeholders will participate in the required stakeholder workshop.

17. Constraints

- The project must use the quantitative figures provided in the case study and must not invent alternative figures.
- The four approval levels supplied in the case study must be used for approval routing analysis.
- The system must address the approval-control weakness demonstrated by the 34 above-limit purchase orders.
- The final academic submission must include the case-study deliverables: SRS, UML Package, Project Plan, Stakeholder & RACI Chart, Test Plan & Evidence, and Risk Register & Closure Note.

18. Dependencies

The case study provides the following project dependencies. Detailed schedule calculations, forward pass, float, critical path and Gantt chart will be documented in the Project Plan.

Activity	Duration	Dependency
Requirements	8 days	None
Approval-limit policy sign-off	6 days	After Requirements
Vendor registration module	14 days	After Requirements
PO module	16 days	After Approval-limit policy sign-off
Vendor acknowledgement	6 days	After PO module
Testing	8 days	After Vendor registration and Vendor acknowledgement
Vendor onboarding drive	15 days	After Testing

19. High-Level Risks

Risk	Probability (P)	Impact (I)
Policy sign-off delayed	0.5	7
Vendors refuse to register online	0.3	6
Approval bypass	0.2	9
Ownership conflict	0.5	6

Risk exposure, ranking, responses, owners and RMMM treatment will be documented in the Risk Register. Problems that have already occurred will be maintained separately in the Issue Log.

20. Stakeholder Communication Requirement

The project shall define what each stakeholder needs to hear, how often the information is required, and from whom it should be communicated. Stakeholders will also be classified by interest and influence. The detailed stakeholder register, interest-influence grid, communication plan and RACI matrix will be included in the Stakeholder & RACI deliverable.

21. Success Criteria

1. Vendor registration can be managed through a centralized portal.
2. Required vendor documentation can be submitted and verified through the defined process.
3. Purchase orders can be created through the system.
4. Purchase orders are routed according to the four approval levels defined in the case study.
5. Approval beyond the applicable authority level is controlled.
6. Vendors can acknowledge purchase orders.
7. Relevant vendor and PO activities are traceable for audit purposes.
8. Sponsor, System Owner and Approver responsibilities are clearly defined.
9. Security, availability and auditability have measurable requirements in the SRS.

10. Every functional requirement is traceable to at least one test case.

22. Business-Level Acceptance Criteria

ID	Acceptance Criterion
AC-01	Vendor registration is available through VendorGate.
AC-02	Required vendor documentation can be submitted through the defined process.
AC-03	Vendor documents can undergo verification.
AC-04	Purchase orders can be created through the system.
AC-05	PO approval routing follows the defined four-level structure.
AC-06	A PO cannot be approved by an authority whose applicable limit is insufficient.
AC-07	Vendors can acknowledge purchase orders.
AC-08	Relevant registration and PO activities are traceable for audit purposes.
AC-09	Stakeholder roles for Sponsor, System Owner and Approver are established.
AC-10	Requirements are traceable to corresponding test cases.

23. MoSCoW Prioritization

23.1 Must Have

- Vendor registration
- Vendor document submission
- Document verification
- Purchase-order creation
- Approval-limit enforcement

- Purchase-order approval
- Vendor acknowledgement
- Auditability
- Security controls

23.2 Should Have

Additional supporting requirements should be confirmed during stakeholder elicitation before being placed in this category.

23.3 Could Have

Convenience and enhancement features should only be introduced after the core scope is confirmed.

23.4 Won't Have – Current Baseline

Features not specified by the case study are excluded from the current baseline unless approved through change control.

24. Traceability Foundation

The BRD establishes the business baseline that will be transformed into detailed SRS requirements, UML designs and test cases. The final Requirements Traceability Matrix will link every functional requirement to one or more test cases.

Example: BR-06 (approval-limit enforcement) → detailed SRS requirement → PO approval UML sequence/state → Boundary Value / Decision Table tests → recorded test result.

25. Deliverables Derived from this BRD

Deliverable	Purpose
SRS Document	Detailed scope, 12 functional requirements, 8 measurable NFRs, constraints, MoSCoW priorities and traceability.
UML Package	Use case, class, sequence, activity and purchase-order state diagrams, with cohesion/coupling justification.
Project Plan	WBS, network diagram, critical path, Gantt chart, milestones and resource allocation.
Stakeholder & RACI Chart	Roles, responsibilities, interest, influence, expectations, communication and engagement actions.
Test Plan & Evidence	Test cases, BVA, equivalence classes, decision table, defect log, DRE and defect density.
Risk Register & Closure Note	Probability-impact analysis, exposure-ranked risks, RMMM for top three risks, issue log and lessons learned.

26. Document-to-Project Relationship

The documentation will be developed as a connected chain rather than as independent assignments:

BRD → SOW → SRS → UML/System Design → Project Plan → Sprint Planner → Application → Test Plan → Risk Register → RTM

The BRD defines the business need and baseline. The SRS will then translate the baseline into detailed software requirements. Those requirements will drive the UML package, project schedule, testing and final application implementation.

27. Conclusion

VendorGate is intended to transform the organization's fragmented vendor onboarding and purchase-order processes into a centralized and controlled digital workflow. The current dependence on email, Excel and paper approvals, together with the audit finding of 34 above-limit purchase orders, establishes the business need for stronger process control and traceability.

This BRD establishes the business baseline for the project without introducing unsupported project figures. The subsequent SRS, UML package, Project Plan, Stakeholder & RACI documentation, Test Plan and Risk Register will be developed from this baseline.